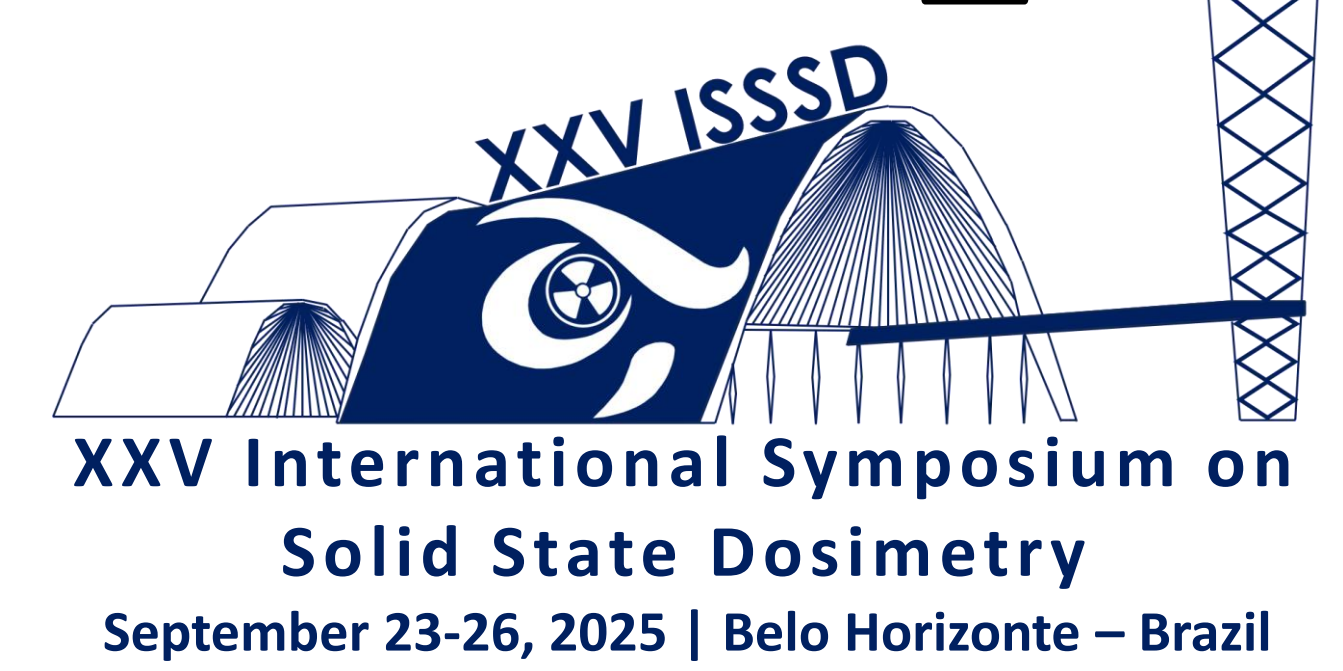


Evaluation of the affect on Quality Control tests of a 3T MRI by the proximity of a subway station and an adjacent MRI scanner

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1. INTRODUCTION

The increasing number of magnetic resonance imaging (MRI) scanners in dense urban settings introduces challenges in mitigating external interferences from nearby large metallic structures. This study evaluates the performance (QC tests) of a 3.0 T MRI scanner since its installation, and assesses potential external interferences resulting from the installation of another MRI scanner in an adjacent room and the construction of nearby subway tracks.

2. MATERIALS AND METHODS

The Magphan 170 phantom was scanned according to ACR standards, specifications in Table 1. A total of 108 MR image acquisitions were collected from the 3T MRI Philips scanner (from its installation in March 2016 until December 2023). QC tests evaluated were: geometric distortion, slice thickness accuracy (STA), signal-to-noise ratio (SNR), percent image uniformity (PIU), percent signal ghosting, low contrast detectability, and high contrast spatial resolution. The Wilcoxon rank-sum test was employed to assess significant changes associated with two interventions: first, in December 2017, following the installation of an additional scanner in an adjacent room; and second, in January 2019, coinciding with the beginning of nearby subway station operations.

Table 1. Sequence parameters used in image acquisitions for quality control tests.

Sequence	Spin Echo
TR (ms)	500
TE (ms)	20
FOV (mm)	256 x 256
MATRIX (pixels)	256 x 256
NEX	1
Bandwidth (Hz/pixel)	364
Flip angle	90
Slice thickness (mm)	5
Time sequence (min)	03:12

3. RESULTS

Table 2 presents the detailed statistical outcomes, including p-values and other numerical findings, obtained from the Wilcoxon rank-sum test.

Table 2. Statistical analysis of QC test results for each period.

QC tests	Period	Mean $\pm$ SD	p-value for interventions	
			1st	2nd
Geometric Distortion (%)	1st	0.58 $\pm$ 0.2	p<0.05	-
	2nd	0.467 $\pm$ 0.149		0.603
	3rd	0.483 $\pm$ 0.187	-	
STA (%)	1st	-1.976 $\pm$ 10.362	0.773	-
	2nd	1.75 $\pm$ 2.688		p<0.05
	3rd	3.967 $\pm$ 2.3	-	
PSG (%)	1st	0.05 $\pm$ 0.027	0.177	-
	2nd	0.04 $\pm$ 0.031		0.252
	3rd	0.044 $\pm$ 0.027	-	
SNR (normalized)	1st	0.359 $\pm$ 0.197	p<<0.05	-
	2nd	0.706 $\pm$ 0.129		0.252
	3rd	0.746 $\pm$ 0.124	-	
PIU (%)	1st	88.229 $\pm$ 4.685	p<0.05	-
	2nd	92.281 $\pm$ 2.521		0.559
	3rd	92.557 $\pm$ 2.796	-	
LCD	1st	34.678 $\pm$ 12.958	0.060	-
	2nd	43.517 $\pm$ 7.936		0.301
	3rd	40.185 $\pm$ 10.77	-	
HCSR	1st	1.014 $\pm$ 0.222	0.817	-
	2nd	0.983 $\pm$ 0.158		0.945
	3rd	0.985 $\pm$ 0.156	-	

4. DISCUSSION

Geometric distortion, SNR and PIU tests presented statistically significant results from the first intervention due to renovation of shielding, which led to better results in all three tests. In the STA test, a significant increase was observed after the beginning of subway operations, which may have been influenced by the gradient field non-uniformity generated by the subway train passing near the MRI scanner, producing eddy currents. Percent signal ghosting, low contrast detectability and high contrast spatial resolution tests did not have significant changes for both interventions.

5. CONCLUSION

The performance evaluation showed that the installation of another MRI scanner in an adjacent room did not interfere. However, geometric distortion, SNR and PIU tests presented better results due to shielding renovation efficiency. Only the STA test showed a statistically significant change following subway construction. This fact may have been influenced by the gradient field non-uniformity generated by the subway train passing near the MRI scanner, producing eddy currents.